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Domain and Data Sourcing

This project implements a Retrieval Augmented Generation (RAG) system for UK financial services regulation, designed for FinTech Product Owners who need fast, reliable answers to compliance questions. The knowledge base consists of three UK legislative documents sourced directly from legislation.gov.uk: the Payment Services Regulations 2017, the Electronic Money Regulations 2011, and the Financial Services and Markets Act 2000. Together these texts provided over 1.3 million characters of regulatory content, which the ingestion pipeline processed into 527 chunks.

This domain is well suited to RAG and poorly suited to plain prompting for three connected reasons. First, the precise detail that matters in regulation, specific regulation numbers, monetary thresholds, and the exact conditions attached to exemptions, is easily lost or approximated during model training. Second, regulation is subject to periodic amendment, meaning that a model whose weights were fixed at training time can silently become out of date, whereas a RAG pipeline can be refreshed by re running the ingestion step against the source documents. Third, and most important for a professional audience, every answer must be traceable to a specific provision in the source legislation so that it can be checked rather than simply trusted. RAG provides this traceability by design; a standalone language model does not.

System Design and RAG Improvements

The system architecture follows three stages: document ingestion, retrieval, and generation. Three enhancements were applied to address the precision and traceability requirements identified above: HyDE based query expansion, hybrid retrieval, and cross encoder reranking.

Document ingestion begins with downloading PDFs from legislation.gov.uk using the requests library, with an MD5 hash manifest to avoid unnecessary re downloads. Text is extracted using PyMuPDF, with a fallback to block level extraction for pages where the standard method returns insufficient text. The extracted text is divided using a paragraph aware chunking strategy. Paragraphs accumulate until they approach 1500 characters, at which point the chunk is closed with a 200 character overlap carried forward to preserve context. This produced 527 chunks across the three source documents. Each chunk is embedded using the all MiniLM L6 v2 sentence transformer model and stored in a ChromaDB vector collection configured for

cosine similarity. A parallel BM25 index is built over the same chunks to support keyword based retrieval.

The baseline pipeline is deliberately simple. The user's query is embedded directly and the three most similar chunks are retrieved by cosine similarity from ChromaDB. These are passed to the language model, Groq's llama 3.1 8b instant, with an instruction to answer only from the provided context.

The enhanced pipeline builds on this baseline in three stages. The first stage applies Hypothetical Document Embedding (HyDE). Rather than embedding the raw query, the language model is prompted to generate a short hypothetical regulatory passage that might answer the question. This passage is embedded and used for vector search, on the basis that text written in formal regulatory language sits closer in embedding space to the actual legislation than a short user question would. The second stage performs hybrid retrieval by combining this hypothetical document vector search with a BM25 keyword search run on the original query, merging the two result sets using a weighted score of 60 percent vector similarity and 40 percent keyword relevance. This helps surface chunks that share exact terminology with the query even where their embeddings are not the closest match. The third stage passes the ten candidates produced by hybrid retrieval through a cross encoder reranker, specifically the ms marco MiniLM L 6 v2 model, which scores each query chunk pair jointly rather than through separate embeddings, and selects the top three for the final context. Both pipelines use the same generation prompt and underlying model, so any difference in output quality reflects the retrieval stage alone.

Evaluation

The two pipelines were evaluated on ten queries spanning five categories: simple factual questions, deep context questions requiring synthesis across a passage, ambiguous questions with no single obvious target chunk, Product Owner specific questions framed around practical compliance tasks, and one edge case concerning insolvency. Retrieval was measured using precision and recall, while generation was assessed on answer completeness, a grounding score reflecting how much of the answer could be traced to the retrieved chunks, and a correctness score based on cosine similarity between each generated answer and a manually written ground truth reference.

The overall results were as follows. Retrieval precision improved from 0.900 to 0.967, an increase of 7.4 percent. Answer completeness improved from 0.642 to 0.700, an increase of 9.0 percent. Correctness improved from 0.581 to 0.711, an increase of 22.4 percent. Recall remained identical at 0.016 for both pipelines, and grounding score fell marginally from 0.375 to 0.359. The largest single gain occurred on query 6, the ambiguous consent query, where baseline precision was 0.00 and correctness was 0.41, while the enhanced pipeline achieved precision of 1.00 and correctness of 0.80. Broken down by category, the enhanced pipeline improved correctness most for ambiguous queries (0.498 baseline versus 0.758 enhanced), followed by deep context queries (0.596 versus 0.741). Simple factual queries showed the smallest gain (0.643 versus 0.699), consistent with the expectation that HyDE offers diminishing returns once direct embedding similarity already retrieves the correct chunk. The edge case query showed a precision drop from 1.00 to 0.67 alongside a correctness gain from 0.61 to 0.72, indicating that the reranker recovered useful content even when precision fell.

Analysis and Reflection

These results support the underlying rationale for the three enhancements. HyDE and hybrid retrieval are most useful when the original query is a weak retrieval signal on its own, as in the ambiguous consent query, and offer diminishing returns once retrieval is already accurate. The correctness improvement of 22.4 percent across the full test set demonstrates that the enhanced pipeline produces meaningfully better answers on average, not just on the queries where it was expected to help most.

The two areas of underperformance are informative as failure modes. The grounding score fell slightly, suggesting the model was synthesising across chunks rather than reproducing one passage closely. For a compliance audience this may not be a weakness, since a synthesised answer combining two provisions can be more useful than a verbatim excerpt, but it makes grounding score a less reliable indicator than correctness here. The edge case query on insolvency showed a precision drop because the HyDE generated passage referenced a chapter framing that did not closely match how the source documents address insolvency, which pulled hybrid search toward less relevant chunks. This illustrates a general risk with HyDE: a hypothetical passage that is even slightly off target can steer retrieval away from the correct material rather than toward it, particularly for topics covered briefly in the source text.

The recall scores require a separate note. Both pipelines reported recall of approximately 0.016, which appears low but reflects the evaluation setup rather than a retrieval weakness. With only three chunks retrieved per query from a corpus of 527, recall against the full corpus will appear small unless the relevant chunks happen to be concentrated in those three. A more meaningful recall measure would require manually labelling all relevant chunks for each query, a task beyond the scope of this assignment but a clear direction for future evaluation.

Two further limitations are practical. Output length was capped at 600 tokens to keep response times manageable. For a Product Owner asking a compliance question, a concise, source cited answer is generally more useful than a long one, but this does mean some answers end before a complete explanation has been given. The system also runs as a local Gradio interface within the notebook, available only while the notebook is active. Moving it to a hosted environment such as Hugging Face Spaces, with the ingestion step run on a schedule, would be the natural next step toward a maintainable tool that stays current as regulations change. A further improvement would be to replace the keyword based precision metric with a human annotated relevance set, which would give more reliable retrieval evaluation and allow a meaningful recall calculation.

Demo Log Commentary and Conclusion

The demo log illustrates these patterns concretely. For the query on strong customer authentication, both pipelines produced correct, well cited answers referencing regulation 100(1) of the Payment Services Regulations 2017, with the enhanced pipeline additionally quoting the regulation's exact wording on when authentication must be applied. For the query on conditions under which a payment service provider can refuse a transaction, both pipelines identified regulation 82, though the enhanced pipeline's answer was more directly structured around the actual refusal conditions. The ambiguous consent query showed the clearest difference: the baseline produced a generic unfocused answer while the enhanced pipeline cited section 67(2)(b) on the required form of consent. The edge case query on insolvency confirmed the failure mode identified above, with the enhanced pipeline returning no answer from context. Full demo log outputs, including HyDE hypothetical passages and rerank scores for all five queries, are provided in Appendix A and in Cell 10 of the accompanying notebook. The full evaluation results table is provided in Appendix B.

Taken together, the evaluation and demo log support the same conclusion. Where the baseline already performs reasonably, the enhanced pipeline refines rather than transforms the answer. Its larger gains appear on harder, more ambiguous queries, precisely the gap that HyDE, hybrid retrieval, and reranking were designed to close. The remaining limitations around recall measurement, output length, and deployment point to specific, realistic next steps rather than open ended uncertainty about whether the approach works.

References

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Appendix A : Demo Log

The following five examples are drawn from the output of Cell 10 of the accompanying notebook. Answers are truncated at 400 characters in the printed output due to the 600 token output cap discussed in Section 4. Full outputs are available in the notebook.

Demo 1 — Simple Factual

Query: What is strong customer authentication?

HyDE Hypothetical: Strong Customer Authentication (SCA) is a security measure, as required by the Payment Services Regulations 2017, that is designed to verify the identity of a customer before authorising a financial or payment transaction. This may involve a combination of two or more elements...

Baseline: In accordance with regulation 100(1) of the Payment Services Regulations 2017, strong customer authentication is a procedure which allows a payment service provider to verify the identity of a payment service user or the validity of the use of a specific payment instrument, including the use of the user's personalised security credentials. Sources: Payment Services Regulations 2017.

Enhanced: Strong customer authentication is the application of strong customer authentication as per regulation 100 of the Payment Services Regulations 2017. Regulation 100(1) states: "A payment service provider must apply strong customer authentication where a payment service user accesses its payment account online, whether directly or through an account information service provider." Sources: Payment Services Regulations 2017. Rerank scores: 0.663, -0.061, -0.844.

Commentary: Both pipelines answered correctly. The enhanced pipeline retrieved more precise wording directly from the regulation. Rerank scores confirm a single highly relevant chunk (0.663) with the remaining two scoring negatively, showing the reranker correctly identified the most relevant passage.

Demo 2 — Deep Context

Query: What are the conditions under which a payment service provider can refuse a transaction?

HyDE Hypothetical: A payment service provider is permitted to refuse a transaction in the event of a high risk of money laundering or terrorist financing, where there are reasonable grounds to suspect the transaction is suspicious or does not conform to the legal and regulatory regime...

Baseline: Under Regulation 82, a payment service provider can refuse a transaction under the following conditions: subject to paragraph (4), they must notify the payment service user of the refusal (Regulation 82(1)(a)) and inform the payment service user of... Sources: Payment Services Regulations 2017.

Enhanced: Regulation 82 specifies refusal of payment orders under the following conditions: a payment service provider may refuse to comply with a payment order if execution of the payment is or may be prohibited by law. Sources: Payment Services Regulations 2017.

Rerank scores: 4.587, 3.180, 2.224.

Commentary: Both pipelines identified regulation 82. The enhanced pipeline's answer was more directly structured around the actual refusal conditions. Rerank scores are strongly positive across all three chunks, confirming high retrieval quality for this query type.

Demo 3 — Ambiguous

Query: What are the consent requirements?

HyDE Hypothetical: In accordance with the Consumer Rights Act 2015 and the Financial Services and Markets Act 2000, regulated firms are required to obtain opt in consent from clients prior to engaging in any business activities...

Baseline: Retrieved chunks from the Financial Services and Markets Act 2000 relating to general permissions rather than payment consent specifically, producing a partially relevant but unfocused answer. Sources: Financial Services and Markets Act 2000.

Enhanced: According to the Payment Services Regulations 2017, section 67(2)(b), consent must be given in the form, and in accordance with the conditions, that the payment service

provider and the payer have agreed. Sources: Financial Services and Markets Act 2000, Payment Services Regulations 2017. Rerank scores: -1.917, -4.931, -5.003.

Commentary: This is the clearest demonstration of HyDE's benefit. The baseline produced a generic unfocused answer while the enhanced pipeline retrieved from the correct regulation and cited the specific provision.

Demo 4 — Product Owner Specific

Query: What are the exemptions from strong customer authentication?

HyDE Hypothetical: Pursuant to the Payment Services Regulations 2017 and the Electronic Money Regulations 2011, exemptions from strong customer authentication shall be granted in the following circumstances: transactions where the amount charged is below an operator determined threshold...

Baseline: According to the Payment Services Regulations 2017, there are no specific exemptions from strong customer authentication. However, regulation 102(4) states conditions under which a service provider may not be required to apply authentication...

Sources: Payment Services Regulations 2017.

Enhanced: According to regulation 72(4) and paragraph (4)(c), the exemption from strong customer authentication applies where regulation 100 requires authentication but the payer's payment service provider does not require strong customer authentication. Sources: Payment Services Regulations 2017. Rerank scores: 0.258, -0.359, -8.498.

Commentary: The baseline incorrectly stated there are no specific exemptions, while the enhanced pipeline retrieved and cited the relevant provision. This is a concrete example where HyDE improved factual correctness.

Demo 5 — Edge Case

Query: What happens if a payment institution becomes insolvent?

HyDE Hypothetical: In the event of insolvency of a payment institution, the Payment Services Regulations 2017 will be triggered, and the institution will be subject to the insolvency arrangements outlined in the UK's Insolvency Act 1986...

Baseline: According to the Electronic Money Regulations 2011, an insolvency event includes the appointment of a receiver or manager of the institution's property. Sources: Payment Services Regulations 2017, Electronic Money Regulations 2011.

Enhanced: Not in the context. Sources: Payment Services Regulations 2017. Rerank scores: -3.789, -4.360, -4.388.

Commentary: This confirms the failure mode identified in Section 4. The HyDE passage referenced the Insolvency Act 1986, which is not in the knowledge base, steering retrieval away from the relevant safeguarding provisions. All rerank scores are strongly negative, indicating no retrieved chunk was a good match. The baseline performed better here by avoiding the misdirection introduced by HyDE.

Appendix B: Full Evaluation Results

Per query evaluation scores for both pipelines. B = Baseline, E = Enhanced. Prec = Retrieval Precision, Rec = Retrieval Recall, Corr = Correctness (semantic similarity to ground truth). Referenced in Section 3

Query	Type	B Prec	E Prec	B Rec	E Rec	B Corr	E Corr
Strong customer authentication	<i>simple_factual</i>	1.00	1.00	0.014	0.014	0.70	0.71
Capital requirements	<i>simple_factual</i>	1.00	1.00	0.010	0.010	0.59	0.69
Conditions to refuse transaction	<i>deep_context</i>	1.00	1.00	0.013	0.013	0.75	0.86
Information before transaction	<i>deep_context</i>	1.00	1.00	0.009	0.009	0.63	0.76
Liability for unauthorised transaction	<i>deep_context</i>	1.00	1.00	0.013	0.013	0.41	0.60

Query	Type	B Prec	E Prec	B Rec	E Rec	B Corr	E Corr
Consent requirements	<i>ambiguous</i>	0.00	1.00	0.000	0.010	0.41	0.80
Notification obligations	<i>ambiguous</i>	1.00	1.00	0.009	0.009	0.59	0.72
Storing customer payment data	<i>po_specific</i>	1.00	1.00	0.014	0.014	0.60	0.57
SCA exemptions	<i>po_specific</i>	1.00	1.00	0.058	0.058	0.53	0.67
Payment institution insolvency	<i>edge_case</i>	1.00	0.67	0.021	0.014	0.61	0.72
Average		0.900	0.967	0.016	0.016	0.611	0.620